Worksheet 5

Instructions

- Include your full name and Coyote ID on the top right of the first page.
- Submit completed worksheets to Gradescope (no tagging required).
- Submit before 11:59 PM on Wednesday, September 30, 2026

Physics Problem Solving Tips

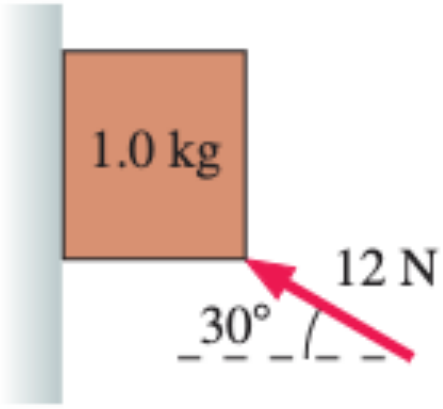
- Draw a diagram (if possible).
- Write down what you know.
- Write down what you want to find.
- First solve symbolically; plug in values at the end.
- Check to make sure your units are correct.
- If you plug in values, check order of magnitude.

Problems

This worksheet covers using Newton's second law (with and without constant velocity) and friction (static and kinetic).

1. Write the names of everyone you worked with.
2. In an electricity experiment, a $1.0\,\mathrm{g}$ plastic ball is suspended on a $60\,\mathrm{cm}$ long string and given an electric charge. A charged rod brought near the ball exerts a horizontal electric force $\vec{F}_{\text{elec}}$ on it, causing the ball to swing out to a 20° angle and remain there. (The ball isn't moving so this system is in mechanical equilibrium, i.e., $\vec{F}_{\text{net}} = 0$, it is also under the influence of gravity in addition to $\vec{F}_{\text{elec}}$)
   a. What is the magnitude of $\vec{F}_{\text{elec}}$?
   b. What is the tension in the string?
3. Compressed air is used to fire a $50\,g$ ball vertically upward from a $1.0\,\mathrm{m}$ tall tube. The air exerts an upward force of $2.0\,\mathrm{N}$ on the ball as long as it is in the tube. How high does the ball go above the top of the tube? Neglect air resistance. (This is telling you that there is constant forces on the ball while in the tube, so we are using that information to find the ball's acceleration in the tube, then use our kinematic equations to find the ball's velocity when it exits the tube, at which point we use our kinematic equations and free fall to determine the height of the ball).

FIGURE 1



4. (Friction) A $1.0\,\text{kg}$ wood block is pressed against a vertical wood wall by the $12\,\text{N}$ force shown in Figure 1. If the block is initially at rest; will it move upward, downward, or stay at rest? (This is asking if the applied force is enough to overcome the max static friction force)